



# Medical Microbiology

## Lecture Four

# Immune System

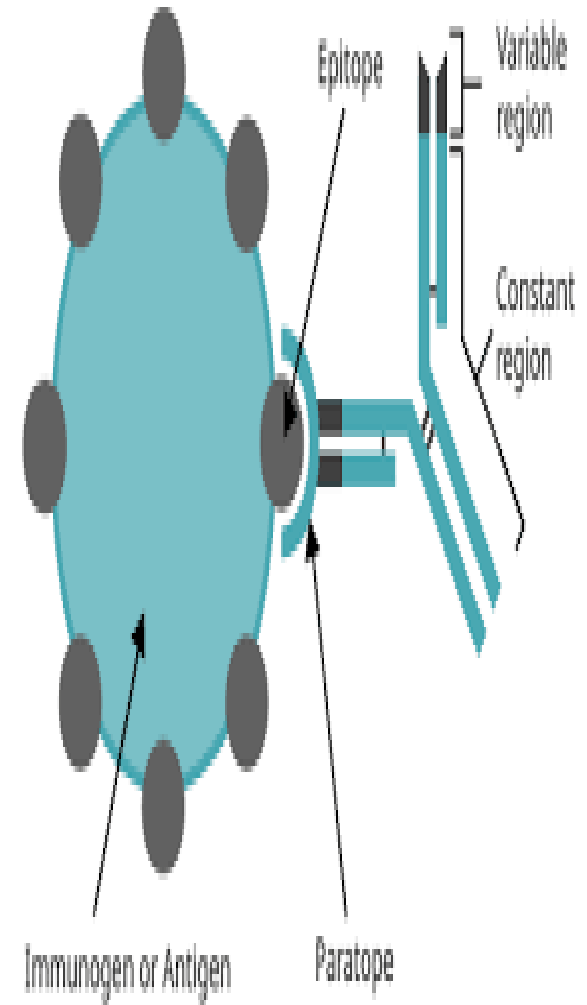
Third stage- College of Dentistry

**Dr. Dheyaa Naji Hamza**

[Dheyaa.naji@alkafeel.edu.iq](mailto:Dheyaa.naji@alkafeel.edu.iq)

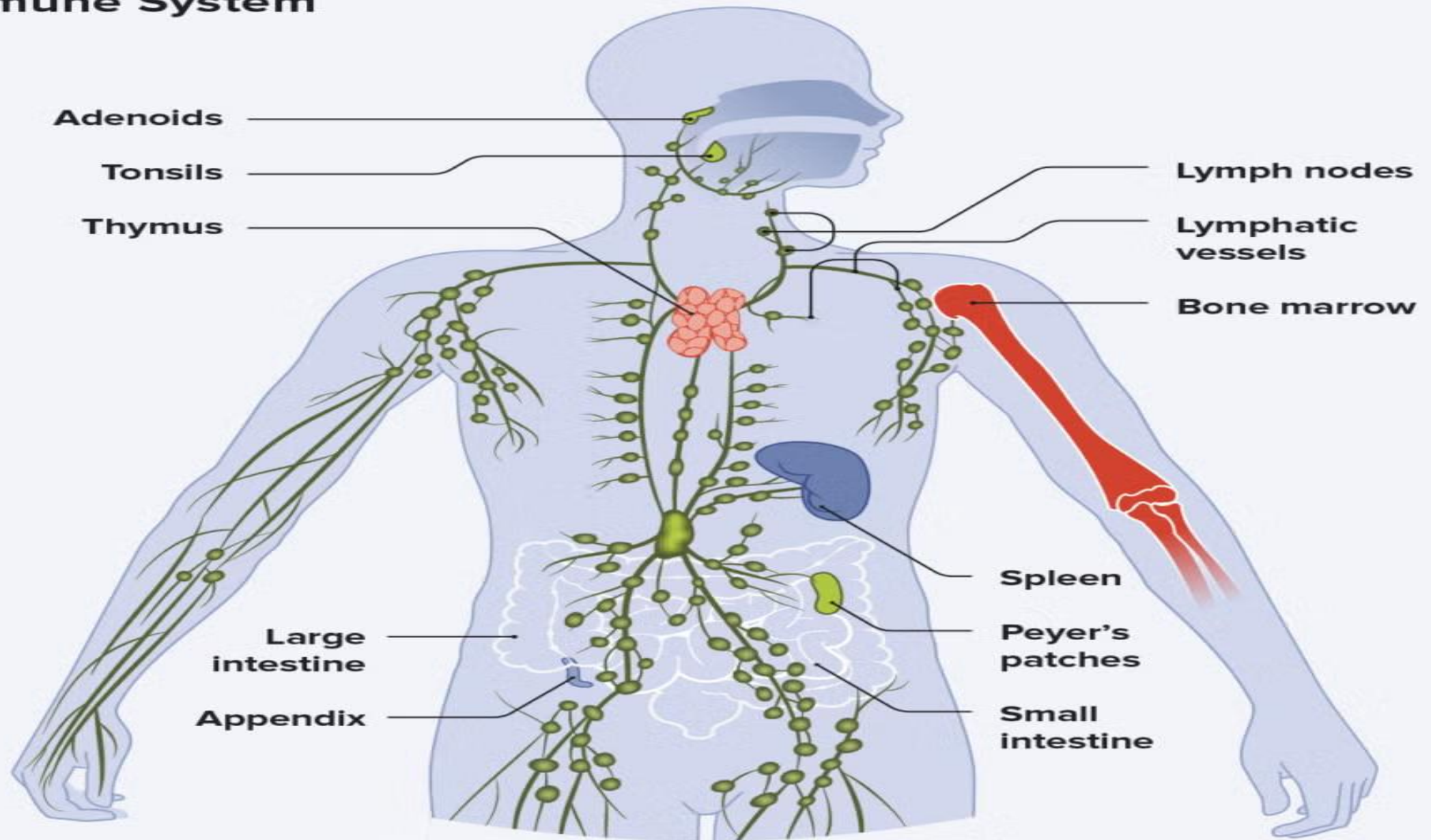
# Introduction: some important terms:

- **Immunogen:** substance that induce specific immune response.
- **Antigen:** substance that react with the product of specific immune response.
- **Antibody:** A specific protein produced by B cells in response to an immunogen, which binds specifically to the corresponding antigen.



- **Adjuvant:** substance that can enhance the immune response to an immunogen.
- **Immunity (or resistance):** refers to the body's ability to defend itself against infectious organisms and other harmful invaders through a sequence of actions known as the immune response.
- **The immune system:** Composed of specialized cells, tissues, organs, and proteins it protects the body against harmful invaders, helping maintain health and prevent infections.

# Immune System



# The Lines that Protection Against Invading Pathogens:

**1. First Line of Defense:** Non-specific natural barriers which restrict entry of pathogen.

**Examples:** Skin and mucous membranes.

**2. Second Line of Defense:** Innate non-specific immune defenses provide rapid local response to pathogen after it has entered host.

**Examples:** Fever, phagocytes (macrophages and neutrophils), inflammation, and interferon.

**3. Third line of defense:** Antigen-specific immune responses, specifically target and attack invaders that get past first two lines of defense.

**Examples:** Antibodies and lymphocytes.

**There are two important types of immune system:**

**A. Innate** (natural) or **non-specific** immunity.

**B. Adaptive** (acquired) or **specific** immunity.

**Innate (Natural) or non-specific immunity characteristic by:-**

- It is present at birth.
- Has the ability to resist infection by means of normally present body functions.
- Many of these mechanisms are subject to influence by many factors such as: nutrition, age, stress, and genetic determinants.

**Innate defenses can be classified into three main groups:**

1. Barriers to infection.
2. Humoral factor (Soluble proteins).
3. Cells.



# 1.Barriers to infection :-

- **Physical barriers:** Skin and mucosal membranes act as physical barriers to the entry of pathogens to body.
- The flushing actions of tears, saliva and urine protect epithelial surfaces from colonization.
- High oxygen tension in the lungs and body temperature, can also inhibit microbial growth.
- In the respiratory tract, mucus is secreted to trap microorganisms.

- **Mechanical barriers:**

Mechanically microorganism expelled by:

- Beating cilia.
- Coughing.
- Sneezing.

## • **Chemical barriers :-**

- The growth of microorganisms is inhibited at acidic pH (e.g. in the stomach and vagina).
- Lactic acid and fatty acids in sebum (produced by sebaceous glands) maintain the skin pH between (3 to 5).
- Enzymes such as lysozyme (found in saliva, sweat and tears) and pepsin (present in the gut) destroy microorganisms.



- **Biological barriers (normal flora):**
- Normal flora develops when non-pathogenic bacteria colonize epithelial surfaces.
- ❖ It helps protect the host by:
  - Competing with harmful bacteria for nutrients and attachment sites.
  - Producing substances with antibacterial properties.

## 2. Humoral factor (Soluble proteins):

- When infectious agents have penetrated tissues another innate defense mechanism play role in protection.
- **Humoral factors** play an important role in inflammations, and these factors found in serum or they are formed at the site of infection, that include:

**1.Complement system:** is a group of serum proteins that are functionally inactive under normal conditions. However, upon activation, they target and damage the membranes of pathogenic organisms, leading to their destruction or aiding in their elimination from the body.

**2. Beta-lysine:** is a protein produce by platelets during coagulation can lyses many bacteria by chemotactic agents for phagocytic cells.

**3. Interferon (INF):** which are group of proteins produced by virus infected cells.

Among the many functions of INF is the ability to bind to nearby cells and induce a generalized antiviral state.

There are three types of them (IFN  $\alpha$ ,  $\beta$ ,  $\gamma$ ).

## 4. Lactoferrin and transferrin:

are iron-binding proteins that compete with microorganisms for iron, an essential metabolite.

**5. Lysozyme:** bactericidal enzyme in mucus, saliva, tears, sweat and breast milk, it cleaves peptidoglycan in the cell wall of microorganisms.

**6. Cytokines:** Tumor necrosis factor, interleukins (IL-1) to (IL-12).

### 3. Cells of innate immunity:

The cells of the innate immune system consist of:

#### A-Phagocytes:

Phagocytes (**macrophages** and **neutrophils**) engulf and then destroy pathogens by process of phagocytosis.

**Macrophages:** are long-lived immune cells found at sites of infection. They release cytokines that attract neutrophils shorter-lived but highly active phagocytic cells.

**Neutrophils:** make up 50–70% of circulating white blood cells. They are among the first responders to inflammation, arriving rapidly at the site, where they kill pathogens and typically die in the process.



## Stages of Phagocytosis:

**1.Chemotaxis:** Phagocytes are chemically attracted to site of infection.

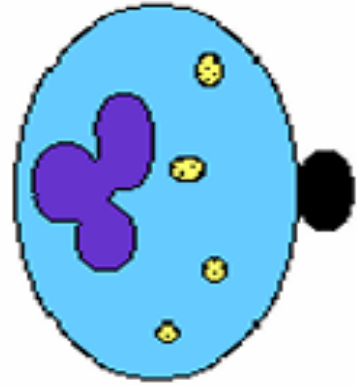
**2.Adherence:** Phagocyte plasma membrane attaches to surface of pathogen or foreign material.

**3.Ingestion:** Plasma membrane of phagocytes extends projections (pseudopods) which engulf the microbe.

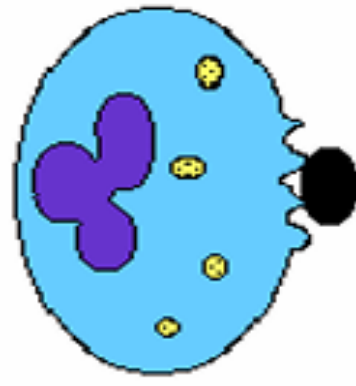
Microbe is enclosed in a sac called **phagosome**.

**4.Digestion:** Inside the cell, phagosome fuses with lysosome to form a phagolysosome.

Lysosome enzymes kill most bacteria within 30 minutes.



**Adherence**



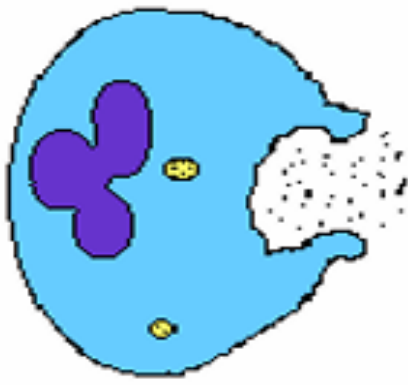
**Membrane activation**



**Phagosome formation**



**Fusion and digestion**

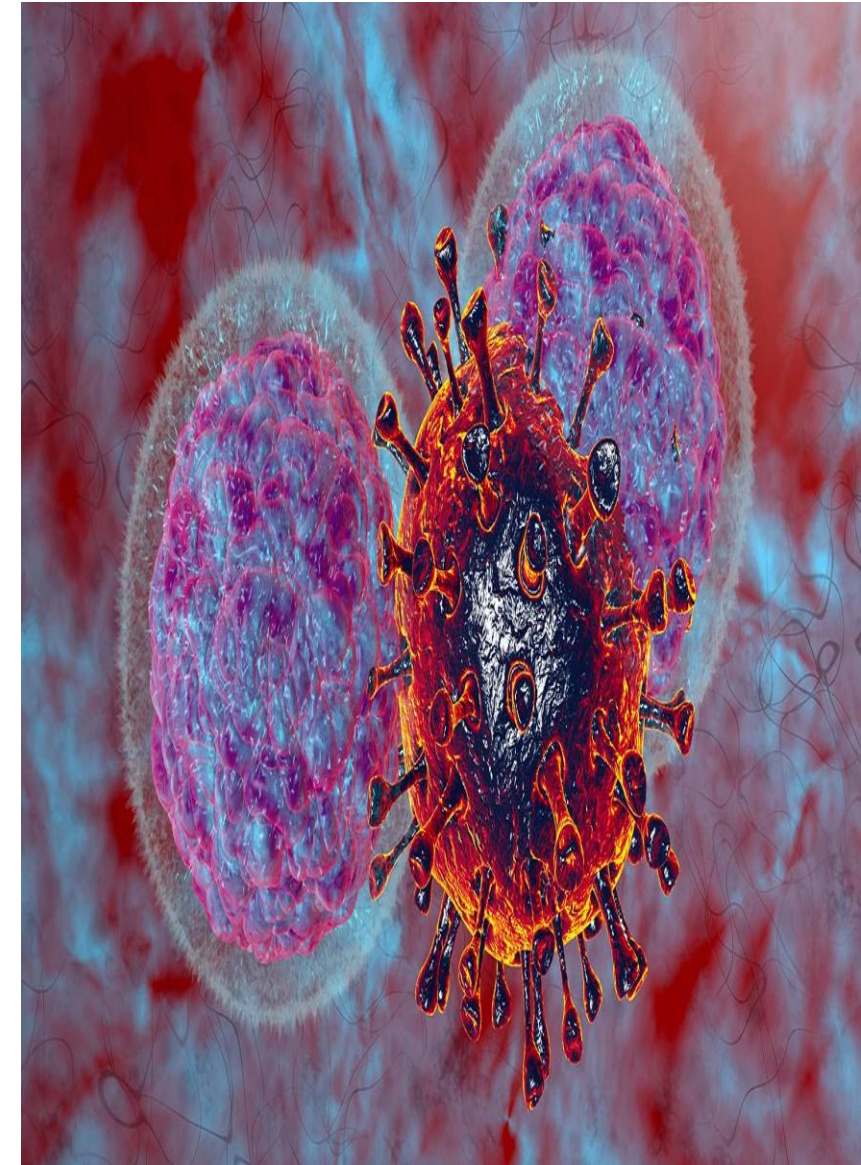


**Release of degraded products**



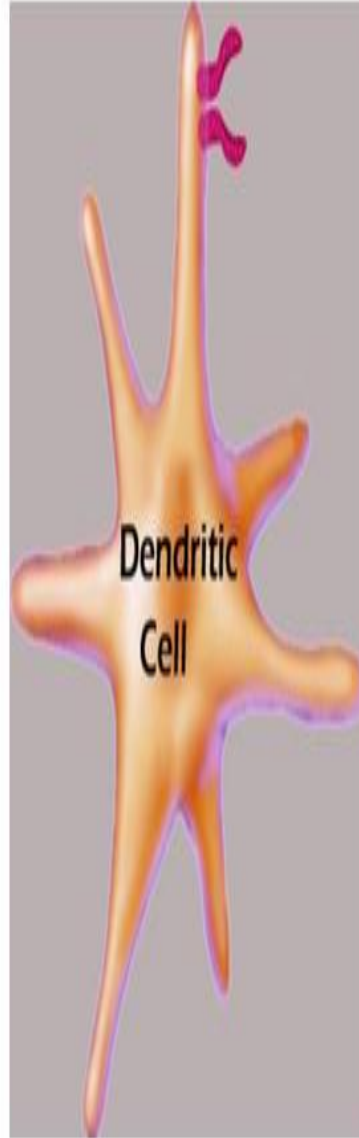
## B-Natural killer (NK) cells:

- Large granular lymphocytes (not B-cell or T-cell) Kills tumor cells & viral infection cells (intracellular pathogens).
- NK cells do not require prior immunization or activation they attach to target cells, then **cytotoxic granules** are released onto surface of cell and effectors cell membrane and induce death.
- **Cytotoxic granules**: which contain key molecules such as:
  - **Perforin** – creates pores in the target cell membrane.
  - **Granzymes** – serine proteases that enter the cell through these pores and induce apoptosis (programmed cell death).



## C-Antigen Presenting Cells (APC):

- **Antigen presenting cells:** are cells that mediate cellular immune response by engulfment, processing and presenting antigens to the T-cell receptor.
  - **Traditional APC include:** Macrophages, **Dendritic cells**, Langerhans cells, and B- lymphocytes.
  - **Dendritic cells** play a unique role because they are involved in both **innate** and **adaptive immunity**:
    - 1-Innate immunity:** They act as first responders by detecting pathogens through pattern recognition receptors.
    - 2-Adaptive immunity:** They process and present antigens to T lymphocytes, activating the adaptive immune response.
- So, **dendritic cells** are a crucial **bridge between innate and adaptive immunity**.



## **Inflammation:**

Triggered by tissue damage due to infection, heat, wound, etc.

### **Major Symptoms of Inflammation:**

1. Redness.
2. Pain.
3. Heat May also observe.
4. Swelling.
5. Loss of function.

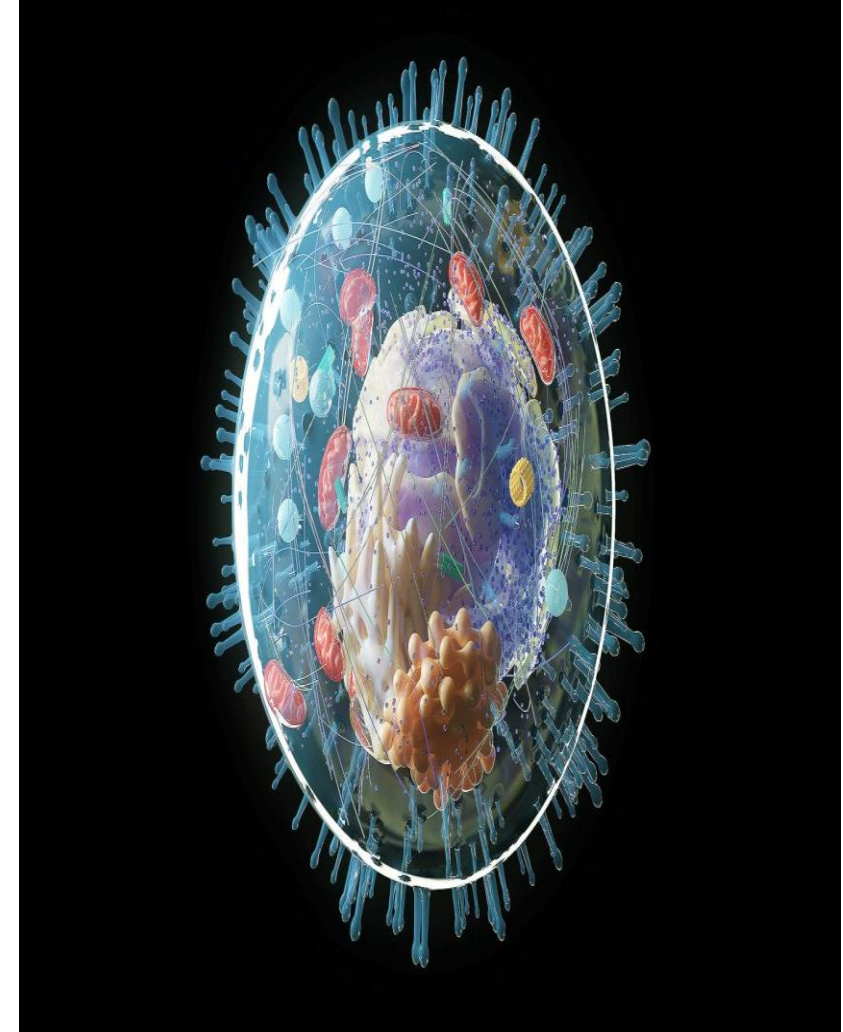
### **Functions of Inflammation:**

1. Destroy and remove pathogens
2. If destruction is not possible, to limit effects by confining the pathogen and its products.
3. Repair and replace tissue damaged by pathogen and its products.



## B. Adaptive or (acquired/specific) immunity:

- **Adaptive immunity** is driven by B and T lymphocytes and is activated upon exposure to infectious agents. It provides a targeted response against specific pathogens and is marked by its ability to recognize and remember previous encounters by **memory cell** that enables a faster and more effective immune reaction upon subsequent exposures to the same pathogen.



**Adaptive immunity is often sub-divided into two major types:**

**a. Naturally acquired immunity:** is occurs through contact with a disease.

**b. Artificially acquired immunity:** is develops only through deliberate actions such as vaccination.

**Adaptive immunity include:**

**1- Humoral Immunity (Antibody Immunity):-**

- Type of immunity that is mediated by secreted antibodies (Ab) produced by the **B-lymphocyte** cells.
- Secreted antibodies bind to antigens on the surfaces of invading microbes (such as viruses or bacteria), which exposure them for destruction.

## 2-Cell-mediated immunity (Cellular Immunity)(T-Cell):

The main types of T lymphocytes (T cells) are:

### 1-Helper T cells (CD4<sup>+</sup> T cells):

- Assist other immune cells by releasing signaling molecules called cytokines.
- Activate B cells, cytotoxic T cells, and macrophages.

### 2-Cytotoxic T cells (CD8<sup>+</sup> T cells):

- Directly kill infected or cancerous cells by inducing apoptosis.

### 3-Regulatory T cells (Tregs):

- Suppress immune responses to maintain immune system balance and prevent autoimmune reactions.

### 4-Memory T cells:

- Long-lived cells that “remember” past infections and respond faster upon re-exposure to the same antigen.

# The innate and adaptive immune response

| Characteristics                                                                                                                                      | Cells                                                                                                                                        | Molecules                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Innate immunity</b>                                                                                                                               |                                                                                                                                              |                                                                                                                                        |
| <ul style="list-style-type: none"> <li>Responds rapidly</li> <li>No memory</li> <li>No specificity</li> <li>No prior exposure is required</li> </ul> | <ul style="list-style-type: none"> <li>Physical barriers</li> <li>Phagocytes (PMNs and macrophages)</li> <li>Natural killer cells</li> </ul> | <ul style="list-style-type: none"> <li>Humoral factors</li> <li>Complement</li> <li>Acute phase Proteins</li> <li>Cytokines</li> </ul> |
| <b>Adaptive immunity</b>                                                                                                                             |                                                                                                                                              |                                                                                                                                        |
| <ul style="list-style-type: none"> <li>Responds Slowly</li> <li>Memory</li> <li>Highly specific</li> <li>Present after exposure to an Ag</li> </ul>  | <ul style="list-style-type: none"> <li>T cells</li> <li>B cells</li> <li>Dendritic cells</li> </ul>                                          | <ul style="list-style-type: none"> <li>Antibodies</li> <li>Cytokines</li> <li>Granzymes</li> </ul>                                     |





**Any Question?**



**Thank you**